

Job Thing

Architecture, data model, API, device constraints and test strategy

JobThing



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Status

Implemented

Version

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Demo

jobthing.vercel.app

1 Overview

Three pieces share one codebase. A **host server** (Python standard library only, no third-party packages) fetches and ranks postings and stores them in SQLite. A **kiosk UI** (plain JavaScript, no build step) runs in the Car Thing's Chromium and talks only to the server. A **static web demo** runs the same UI in an emulator page, with a shim that answers the API from a snapshot. Requirements are in the PRD; this document covers how they are met.

```

Car Thing (Chromium 69 kiosk, 800x480)
| HTTP over USB ethernet (RNDIS)
| 192.168.7.2 -> host 192.168.7.1:8191
v
Host server (python3 server/server.py) -----+
|- refresh: ThreadPool(6) over sources, hourly
|   greenhouse lever ashby smartrecruiters career sites adzuna remoteive
|- matching + descriptions + ranking
|- SQLite (WAL) data/jobthing.db
|- logo proxy + disk cache (the device never reaches the internet)
|- devtools: screenshots via the kiosk's own Chromium
|- apply/: tailor resume, fill form in Chrome, stop before Submit

Web demo (Vercel, static)
kiosk UI + shim.js
answers /api/* from data/index.json
^ built daily by tools/build_demo.py

```

2 Design decisions

Decision	Why
Standard library only on the server	Runs on any Mac or Linux box with Python 3; nothing to install, nothing to patch. Node and coverage.py are needed only for tests.
No build step for the kiosk	The target is Chromium 69. Writing ES2017 by hand avoids transpiler surprises and keeps the payload small on a 512 MB device.
Server does all networking	The device stays offline. Logos are proxied and cached, so a slow CDN never stalls the dial.
Match on titles, not descriptions	Descriptions mention testing in every engineering role. Titles, with include and exclude patterns, separate QA work from "quality" in manufacturing or food safety.
Scores stored with a basis hash	Ranking version, resume hash and content hash decide when to recompute, so editing a pattern re-ranks without a manual version bump.
One UI, two backends	The demo reuses <code>web/</code> unchanged and swaps the server for <code>shim.js</code> , so the demo cannot drift from the device.

3 Server modules

Modules are layered; each imports only modules above it.

Module	Responsibility
config	Paths, environment settings (JT_PORT , JT_REFRESH , JT_DATA), board list from companies.json merged with an untracked local file.
text , net	HTML to text, remote and employment heuristics, timestamps; HTTPS with a working CA bundle.
db	Connection, schema and migrations, settings, upsert.
matching	Role families, 20 location groups, level from title, company keys. Registered as SQL functions for faceting.
descriptions	Posting HTML to readable blocks, skills, salary bounds.
catalog , facets	Which sources exist and are enabled; faceted counts per preset with a small cache.
sources/	One fetcher per source type, each returning normalized rows.
refresh	Parallel fetch, per-source cadence (every_hours), location filters (only), pruning of closed postings, status and progress.
ranking , trends	Match score and reasons; per-preset market summary.
logos , devtools	Logo resolution and cache; kiosk screenshots over the DevTools protocol.
api , app	HTTP routes on http.server ; startup and the refresh loop.

4 Sources

Source	Endpoint	Notes
Greenhouse (60)	boards-api.greenhouse.io/v1/boards/{slug}/jobs	Full descriptions, apply links
Ashby (24)	api.ashbyhq.com/posting-api/job-board/{slug}	Pay ranges often included
Lever (5)	api.lever.co/v0/postings/{slug}	
SmartRecruiters	api.smartrecruiters.com/v1/companies/{slug}/postings	Details fetched only for role titles; non-tech functions dropped
Career sites	Site-specific JSON	For employers with no public ATS board
Adzuna	Search API, free key	Reaches Workday and iCIMS employers; usage ledger stops before caps
Remotive	Public API	At most every 6 hours

Board APIs list only open postings, so after a **successful** fetch any job no longer listed is deleted (saved jobs are kept). A failed fetch never prunes.

5 Data model (SQLite, WAL)

Table	Contents
jobs	id (source:slug:id), source, company, domain, title, location, remote, employment, url, logo_url, description, description_html, posted_at, seen_at. Index on posted_at.
job_scores	job id, score, reasons (JSON), basis hash, skills (JSON), salary_lo, salary_hi.
saved , dismissed	Job id and time (saved jobs can carry a note).
applications	Apply with Claude state per job: status (queued, tailoring, filling, ready, needs you, failed, applied), detail, resume PDF, result JSON, cost.
settings	Key and JSON value: presets, order, source switches and status, Adzuna usage.

6 HTTP API

Route	Purpose
GET /api/jobs	Feed for a preset or explicit role, loc, level, company; <code>order=match new</code> ; paginated.
GET /api/job	One posting: blocks, skills, reasons, other locations.
GET /api/filters , /api/facets	Presets, options and live counts per option.
GET /api/trends	Market summary for preset 1 to 4.
GET /api/sources , /api/health	Per-source health and counts; server status.
GET /api/logo	Cached company logo, content type sniffed.
POST /api/save , /api/dismiss	Save or dismiss a job.
POST /api/presets , /api/order	Store presets and feed order.
POST /api/sources , /api/refresh	Toggle a source; refresh now.
POST /api/open	Open saved postings in the host computer's browser.
POST /api/apply , /api/apply/stop , /api/applications/mark	Prepare applications, stop, mark applied.
POST /api/screenshot	Capture the panel via DevTools.

7 Matching and ranking

Role families (testing, test infrastructure, AI evaluation, hardware test) match the title with include and exclude patterns. At companies flagged as hardware, a bare "Test Engineer" is hardware test. **Location groups** map 800+ free-text variants into Remote, Remote US, US metros, Outside US and regions. **Level** comes from the title: lead, then staff, senior and entry, otherwise mid.

Score component	Points	Rule
Title fit	40	Core SDET and QA engineering titles highest, then test infrastructure, AI evaluation, hardware test
Skills	30	Tools the posting names that the owner's master resume also names
Seniority	15	Senior, staff and lead up; intern and new grad down
Location	10	Remote US and the home metro first
Freshness	5	Added at query time, since it changes hourly

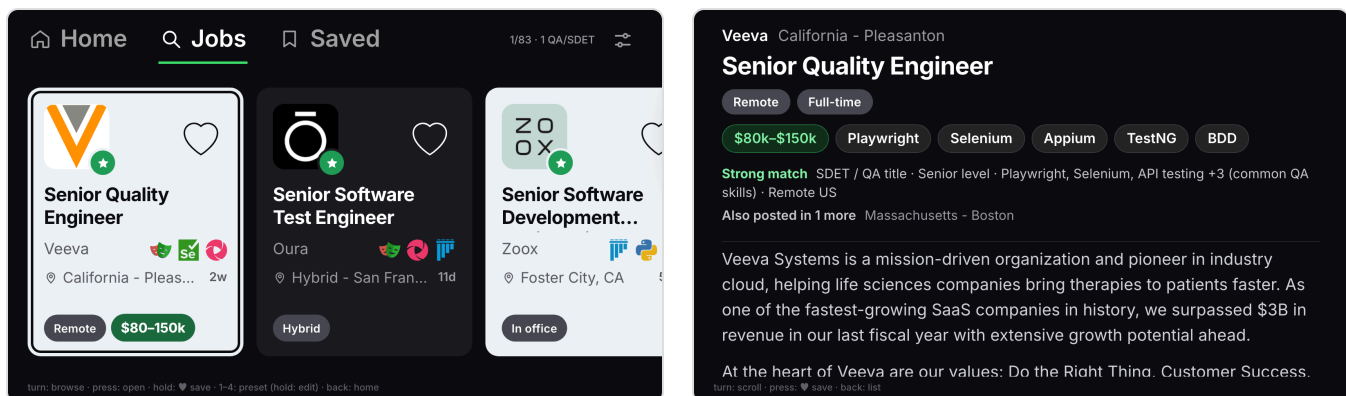
Scores of 80 and above get the strong-match star. Every card can state its reasons, because each component is a named rule rather than a model output.

8 Kiosk UI and device performance

Device fact	Handling
Chromium 69.0.3497	No <code>?.</code> , <code>??</code> , <code>replaceChildren</code> , <code>flexbox</code> <code>gap</code> or <code>inset</code> . Enforced in review and by running the UI tests in Chromium.

Dial emits REL_HWHEEL	Direction read from <code>wheel.deltaX</code> ; events closer than 45 ms are coalesced into one detent.
Dial press is KEY_ENTER	Short press opens or saves; a 600 ms hold saves from the list.
512 MB, earlyoom kills Chromium first	Server-side pagination; stock Spotify app stopped (used memory 213 MB to 133 MB).
60 Hz LVDS panel	Posting text glides with a <code>requestAnimationFrame</code> ease-out (170 ms per detent, whole pixels, <code>will-change: transform</code>).

Detent latency fix. Profiling on the device over DevTools showed each dial detent rebuilt all cards, 272 ms per detent. The rail is now rebuilt only when the job list changes and a detent just moves the selection class: **6 ms per detent**. A UI test asserts the card elements survive a detent, so the regression cannot return silently.



Feed: logo, star for strong match, tool logos, age, work mode and pay. Posting view: tags, match reasons, dial-scrolled text.

9 Apply with Claude

- Tailor.** `claude -p` rewrites the resume toward the posting from a master resume. Code then verifies the result: same employers, titles, dates and contact details; every bullet traced to a master bullet; no number or tool absent from the master; banned claims rejected. Two rejections fall back to the master. PDF rendered by headless Chrome.
- Fill.** `claude -p --chrome`, limited to browser tools, fills the application and uploads the PDF, then stops before Submit. It never signs in, creates accounts, solves CAPTCHAs or answers self-identification questions; those come back as "needs you".
- Review.** The owner checks each tab, submits, and marks the job applied. Each run has a spending cap (tailor \$1, fill \$4).

10 Web demo

`tools/build_demo.py` fetches the public boards, keeps QA-family roles, and writes `data/index.json` (roles, locations, levels, default presets, sources, jobs with skills and pay). Private boards are excluded. Vercel runs the build on every push; a scheduled workflow pushes an empty commit each morning so the snapshot is at most a day old. `demo-src/shim.js` intercepts `fetch` for `/api/*` and keeps each visitor's presets and saved jobs in their own `localStorage`. "Open all on your computer" becomes a printable checklist named by date.

11 Test strategy

Layer	Scope	Runner	Tests
Backend	Matching and descriptions on real titles, locations and HTML; ranking; trends; each source parser; refresh pruning and cadence; Apply checks; API in process	Python unittest	125
API	The HTTP contract the device depends on, from outside the server	Playwright <code>api</code>	59
UI	The kiosk at 800x480, driven like hardware: wheel as dial, Enter as press, Escape as back, 1 to 4 as presets	Playwright <code>ui</code>	
Web demo	Emulator page, boot, controls, shim API, checklist export	Playwright <code>demo</code>	

- **Fixture servers.** Playwright starts the real server on fixture data (never the owner's data) with a fake desktop opener and a DevTools stub, plus a demo built from the same fixture. Test hooks reset state and seed prepared applications.
- **Evidence.** Every UI and demo test records video and a trace. Python coverage (lines and branches) and V8 JavaScript coverage of the kiosk are collected.
- **Regression first.** Bugs found in use become tests before the fix: a POST to an unknown route that corrupted the next keep-alive request, a saved-job sheet taller than the screen, card rows overflowing at larger pay text, and the detent rebuild.
- **CI and reporting.** GitHub Actions runs all layers on every push and daily, and attaches an interactive report to each run (each test described, filterable, with videos, traces and coverage). The job fails if any test fails.

12 Security and privacy

- No third-party server dependencies; the attack surface is the standard library and SQLite.
- Secrets (Adzuna key) and personal data (resume, profile, applications, private boards) live in `data/`, which is untracked.
- The server binds 0.0.0.0:8191 so the device can reach it over USB ethernet; run it on a trusted network. The device itself only talks to the server.
- The demo is static: no backend, no accounts, and visitor state never leaves the browser.

13 Operations

Command	Effect
<code>python3 server/server.py</code>	Start the server on 0.0.0.0:8191 with hourly refresh
<code>bash bin/push.sh</code>	Remount the device read-write, push <code>web/</code> , restart the kiosk
<code>bash bin/spotify-app.sh off on</code>	Stop or restore the stock Spotify app
<code>bash bin/restore-stock.sh</code>	Put the stock web app back
<code>npm test</code> , <code>npm run dashboard</code>	Run all Playwright suites; open the local test dashboard